
Problem Set 2 - Solutions

Problem 2.1

A nonconducting container filled with 25 kg of water at 20°C is fitted with a stirrer which is made to turn by gravity acting on a weight of mass 35 kg. The weight falls slowly through a distance of 5 m in driving the stirrer. Assuming that all work done on the weight is transferred to the water and that the local acceleration of gravity is $9.8 \frac{m}{s^2}$, determine:

- (a) The amount of work done on the water.
- (b) The internal energy change of the water.
- (c) The final temperature of the water, for which $C_p = 4.18 \frac{kJ}{kg \cdot ^\circ C}$.
- (d) The amount of heat that must be removed from the water to return it to its initial temperature.
- (e) The total energy change of the universe because of (1) the process of lowering the weight, (2) the process of cooling the water back to its initial temperature, and (3) both processes together.

SOLUTION

Part (a)

The work done on the water comes from the elevation change of weight.

$$35 \text{ kg} * 9.8 \frac{m}{s^2} * 5 \text{ m} * \frac{1 \text{ J}}{\frac{1 \text{ kg} * m^2}{s^2}}$$

1715. J

The work done on the water is 1715 J. //ANS

Part (b)

“Nonconducting” means well-insulated, which means $Q=0$. The change in internal energy is equal to the amount of work done on the system, since $\Delta U = Q + W = W$.

$$(*\Delta U = Q + W = W = 1715 \text{ J} = 1.715 \text{ kJ} *)$$

The internal energy change of the water is +1715 J. //ANS

Part (c)

$$\text{In[*] := eq1 = 1.715 kJ == 25 kg * } \frac{4.18 \text{ kJ}}{\text{kg * } ^\circ \text{C}} * (T2 - 20 ^\circ \text{C}) ;$$

$$\text{In[*] := Solve[eq1, T2]}$$

$$\text{Out[*] = } \{ \{ T2 \rightarrow 20.01641 ^\circ \text{C} \} \}$$

The final temperature of the water is 20.0164°C. //ANS

Part (d)

$$(*Q = -W = -1715.J = -1.715 \text{ kJ}*)$$

The amount of heat that must be removed from the water is 1.715 kJ. //ANS

Part (e)

Energy is conserved in all cases. The total energy change of the universe is zero. //ANS

Problem 2.4

An electric motor under steady load draws 9.7 amperes at 110 volts, delivering 1.25 hp of mechanical energy. What is the rate of heat transfer from the motor, in kW?

SOLUTION

For a closed system, the first law of thermodynamics is $\Delta U = Q + W$. In an electrical motor, energy enters as electrical power and leaves as heat and mechanical work. So for an electrical motor, the first law looks a little different $\Delta U = Q + P_{\text{electrical}} + W$. Assume $\Delta U = 0$ at steady state since the temperature of the motor is not changing, so $0 = Q + P_{\text{electrical}} + W$.

$$\begin{aligned} \text{In[*]:= } & (*P_{\text{electrical}} = \text{voltage} * \text{current} *) \\ & P_{\text{electrical}} = 9.7 \text{ amp} * 110 \text{ volt} * \frac{1 \text{ W}}{1 \text{ amp} * \text{volt}} * \frac{1 \text{ kW}}{1000 \text{ W}} \end{aligned}$$

$$\text{Out[*]:= } 1.067 \text{ kW}$$

Work, as in work delivered, is -1.25 hp leaving the motor. Convert this to kW:

$$W = -1.25 \text{ hp} * \frac{1 \text{ kW}}{1.34102 \text{ hp}}$$

$$\text{Out[*]:= } -0.9321263 \text{ kW}$$

$$\text{In[*]:= } \text{eq1} = 0 == Q + P_{\text{electrical}} + W;$$

$$\text{In[*]:= } \text{Solve}[\text{eq1}, Q]$$

$$\text{Out[*]:= } \{ \{ Q \rightarrow -0.1348737 \text{ kW} \} \}$$

The rate of heat transfer from the motor is $Q = -0.135 \text{ kW}$. //ANS

Problem 2.9

Heat in the amount of 7.5 kJ is added to a closed system while its internal energy decreases by 12 kJ. How much energy is transferred as work? For a process causing the same change of state but for which the work is zero, how much heat is transferred?

SOLUTION**Part (a)**

Use the first law for closed systems in the form $\Delta U = Q + W$, with ΔU and Q given, to solve for W .

$$Q = 7.5; \text{ (*kJ*)}$$

$$\Delta U = -12; \text{ (*kJ*)}$$

$$W = \Delta U - Q \text{ (*kJ*)}$$

Out[*]=

$$-19.5$$

-19.5 kJ of energy is transferred as work out of the system (19.5 kJ of energy is transferred to the surroundings). //ANS

Part (b)

Use the first law for closed systems in the form $\Delta U = Q + W$ with ΔU and W given to solve for Q . Since $W=0$, the amount of heat transferred is equal to the change in internal energy.

$$\Delta U = -12; \text{ (*kJ*)}$$

$$W = 0; \text{ (*kJ*)}$$

$$Q = \Delta U - W \text{ (*kJ*)}$$

Out[*]=

$$-12$$

-12 kJ of energy is transferred out of the system as heat. //ANS

Problem 2.11

An incompressible fluid ($\rho = \text{constant}$) is contained in an insulated cylinder fitted with a frictionless piston. Can energy as work be transferred to the fluid? What is the change in internal energy of the fluid when the pressure is increased from P_1 to P_2 ?

SOLUTION

An incompressible fluid cannot be compressed in a piston, so $W = F \cdot dl = 0$. //ANS Since the cylinder is insulated, assume $Q = 0$. From the first law, $\Delta U = Q + W = 0$. //ANS

Problem 2.13

An electric motor runs “hot” under load, owing to internal irreversibilities. It has been suggested that the associated energy loss can be minimized by thermally insulating the motor casing. Comment critically on this suggestion.

SOLUTION

Insulation and heat transfer are at the system boundary and do not affect irreversibilities that are internal to the system. Therefore, energy losses due to internal irreversibilities are not affected by insulation. //ANS